

GUIDEBOOK OF
PALO DURO CANYON



***West Texas State
University
Geological Society***

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PALO DURO CANYON



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*West Texas State University Geological
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DEPARTMENT OF GEOSCIENCES

The Department of Geosciences (geology, geography, anthropology) is housed in the Science Center on the campus of West Texas State University. Additional departmental space is found in the Killgore Research Center and Old Main. The Department offers a program of study leading to a Bachelor of Science and Bachelor of Arts degrees in geology and geography and a Bachelor of General Studies degree in anthropology. Most students are enrolled in the Bachelor of Science in geology degree, designed to meet the needs of students preparing for a professional career in geology. It requires a minimum of 39 semester hours of geology, plus supporting coursework in other sciences, mathematics, and technical writing or cartography. The Department also offers a Master of Science degree in geology. Recent thesis topics include *Geology of Fortress Cliff Quadrangle, Randall County, Texas* and *Sedimentology and Petrology of the Javelina Formation, Big Bend National Park, Brewster County, Texas*.

Members of the departmental faculty have a wide range of academic interests and come from a variety of colleges and universities. Special interests of the faculty include stratigraphy, biostratigraphy, geomorphology, sedimentology, structural geology, tectonics, igneous and metamorphic petrology, petroleum geology, paleontology, cartography, archaeology, and Indians and their culture of the south central United States. In addition, the Department retains a broad concern for earth-science education, and offers courses in introductory earth science and geology to meet student needs in the University general education and in teacher education.

The Department supports students with teaching and graduate assistantships, undergraduate laboratory assistants, and scholarships. Information regarding degree programs and financial aid can be obtained from the Department of Geosciences, West Texas State University, Box 938, Canyon, Texas 79016 or by calling the departmental office at 806-656-2581.

FOREWORD

The West Texas State University Geological Society was organized in 1958 by students of the Department of Geology. The objective of the Society is to promote interest in geology as an academic subject and as a professional career. In order to present the concepts of geology to interested groups, the Palo Duro Canyon Guidebook is sponsored by the WTSU Geological Society.

The Geological Society is indebted to Professor Jack T. Hughes and to Mr. Jerry Harbour for their work in the first edition of this guidebook.

DO YOU HAVE A GROUP OR KNOW OF A GROUP WHO WOULD LIKE A GUIDED FIELD TRIP OF PALO DURO CANYON? CONTACT THE WEST TEXAS STATE UNIVERSITY GEOLOGICAL SOCIETY AT 656-2581 FOR INFORMATION.

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INTRODUCTION

Palo Duro Canyon State Park is located 13 miles east of Canyon, Texas, on State Highway 217 and 17 miles southeast of Amarillo, Texas. The park encompasses about 15,000 acres of eastern Randall and western Armstrong counties.

The initial park area was purchased by the State of Texas in 1931. In 1973 the park boundary was extended to incorporate a famous topographic structure, the Lighthouse (frontispiece). Excellent picnic and camping facilities are available within the Park.

Extending away from the canyon rim is a gently undulating land surface called the Llano Estacado or Staked Plains. It is part of the High Plains, a vast piedmont plain which extends along the eastern base of the Rocky Mountains from Wyoming to Texas. The eastern edge of this plain is, in places, an abrupt escarpment (cliff) known as the caprock. Palo Duro Canyon is a westward extension of this escarpment that has been carved into the High Plains by the Prairie Dog Town Fork of the Red River.

In the park area the canyon is several miles wide. The canyon rim is about 3,500 feet above sea level and the canyon floor, although highly irregular is approximately 2,700 feet above sea level. The maximum depth of the canyon is about 800 feet. The United States Geological Survey has published an excellent topographical contour map of the canyon, the Fortress Cliff Quadrangle. It can be purchased at the park or from the United States Geological Survey.

The Park area normally receives about 20-30 inches of rainfall per year and has a frost free period of approximately 200-240 days per year. The yearly temperature ranges from 0-70°F in the winter and from 65 to 100°F in the summer. The weather is considered fair about 75% of the time. The nights are cool even in the summer.

ECOLOGY

Palo Duro Canyon is part of the escarpment system that forms the eastern boundary of the Llano Estacado or Staked Plains. The Staked Plains in this area is a nearly-level to gently-rolling terrain covered with a mantle of calcareous loess (a wind-blown silt) that has given rise to deep soils with a clay-loam surface and a clay subsoil. These soils are ideally suited to the growth of short grasses, especially blue grama and buffalo grass. Mesquite, yucca, prickly pear cactus, and forbs are common invaders of overgrazed areas.

Many shallow lake basins (playas) dot the plains with a flora different from the surrounding areas. In the playas, plains grasses are displaced by forbs that are suited to withstand the prolonged flooding within the shallow basins.

Near the canyon, the deep soils can be seen grading into shallow, grayish-brown, gravelly-loam soils. The abundance of grasses decreases and mid grasses such as little bluestem and sideoats grama increase. The mid grasses are better able to absorb nutrients from the less fertile soils near the canyon rim than are the short grasses. Especially conspicuous are increasing 3 numbers of shrubs, particularly the evergreen and scale-leaved junipers. Just along the rim of the canyon, the shrubby mountain mahogany is found.

The rugged terrain of the sides of the canyon, extending from the Ogallala downward through the Trujillo and Tecovas formations to the upper part of the Quartermaster Formation, shows a variety of soil types. On the steeper slopes, plants are unable to gain a foothold as erosion removes soil material as fast as it is formed. On less-steep areas, the well-drained escarpment soil is suited for the development of scarp woodland. The deep, woody roots of trees and shrubs are better able to obtain the deeply infiltrating moisture from these soils than are the shallow and fibrous roots of grasses. The common plants on the level areas are junipers, squaw-bush, and little-leaved sumac. On the drier slopes, feather peabush, catclaw, and salt-bush are

found. Groves of oak occur, but not in the abundance found along the escarpment further to the south.

Below the canyon slopes and extending to the creek are a wide variety of soils and a great diversity of plants. Most of the plants of the plains and escarpment are found here. Some of the soils of the nearly level areas are deep, high in fertility, and hold large amounts of water. Tall grasses, such as indiangrass and switchgrass, occur admixed with mid grasses; a rank growth of vine-mesquite grass often occurs in the areas where runoff water collects; and alkali sacaton grass grows on saline soils. Other trees and shrubs include hackberry, soapberry, wafer ash, button bush, foresteria, and Texas buckthorn. Along the creek, cottonwoods, willows, and salt cedar are common.

Because of the varied topography, diversity of plant life, and 4 geographical locations, the canyon affords an ideal habitat for wildlife. Some of the mammals that occur here are the coyote, porcupine, jackrabbit, cottontail rabbit, raccoon, opossum, ringtailed cat, striped skunk, gray fox, white-footed mouse, woodrat, and bat. Mammals that were once common but are now absent or extremely rare are the bison, black bear, black-footed ferret, lobo wolf, cougar, and bobcat. The moose and American elk were introduced into the canyon but are no longer to be found. White-tailed deer, mule deer, and aoudad sheep have also been introduced and are still present. The mule deer is the most common. A great many types of birds are found either as residents of or migrants to the canyon. A few are the golden eagle, red-tailed hawk, sparrow hawk, Mississippi kite, turkey vulture, blue quail, killdeer, nighthawk, roadrunner, red-headed woodpecker, golden-fronted woodpecker, canyon wren, mockingbird, robin, cardinal, meadowlark, Bullock's oriole, painted bunting, white-crowned sparrow, and lark sparrow.

HISTORY OF MAN IN THE AREA

Archeological studies indicate that the earliest known inhabitants of Palo Duro Canyon lived in the canyon from about 10,000 to 5,000 B.C. These early men hunted bison and now-extinct elephant-like mammoths that roamed the area during the Pleistocene Ice Age. Their stone weapons and artifacts have been found in the canyon. Presumably these primitive people, like those who came later, were attracted by streams and springs in the canyon, and by game that came to feed there. Rock exposed in the canyon provided material for tools and weapons.

Through the centuries, various tribes of Plains Indians, including Apache, Cheyenne, Arapahoe, Kiowa, and Comanche, made use of the canyon as a camping ground. After the arrival of the white man, the canyon became a favorite resting place for buffalo hunters and Indian traders who travelled the Plains. White men first established residence there in 1876.

The last Indian battle in Texas was fought in the canyon south of the Park. Col. Ranald Mackenzie and his raiders, on September 25, 1874, attacked a large encampment of Comanche, Cheyenne, Kiowa, and Arapahoe in the canyon. They destroyed about 100 lodges and 1400 horses and mules. The damage inflicted was severe and by the following spring most of the Indians were returned to the reservation in Oklahoma.

GEOLOGIC HISTORY

The age of the earth is calculated in terms of billions of years. For convenience, geologic time is divided into units, called PERIODS, of different lengths ([Figure 1](#)). Sediments deposited during each period and the fossilized remains of animals and plants found in these sediments give a partial record of the events and life of that period.

A nearly homogeneous (uniform composition) rock layer may be identified as a FORMATION. Formations are usually spread over a wide area like a large blanket and are stacked on top of each other with the oldest at the bottom and the youngest at the top. When viewed in the walls of the canyon, they resemble a huge layer cake and may be traced along the canyon walls.

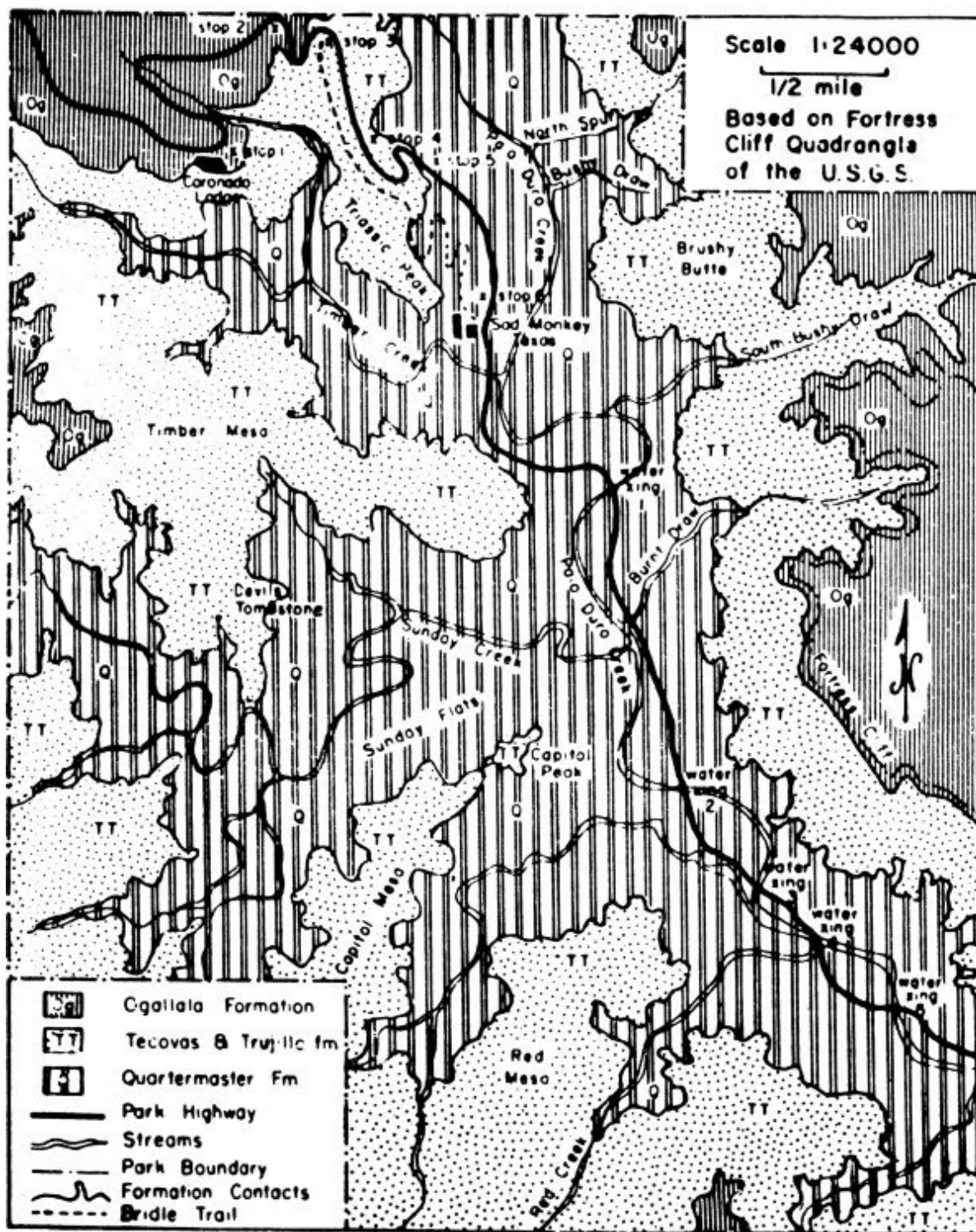


FIGURE 1. GEOLOGIC MAP OF PALO DURO CANYON
[High-resolution Map](#)

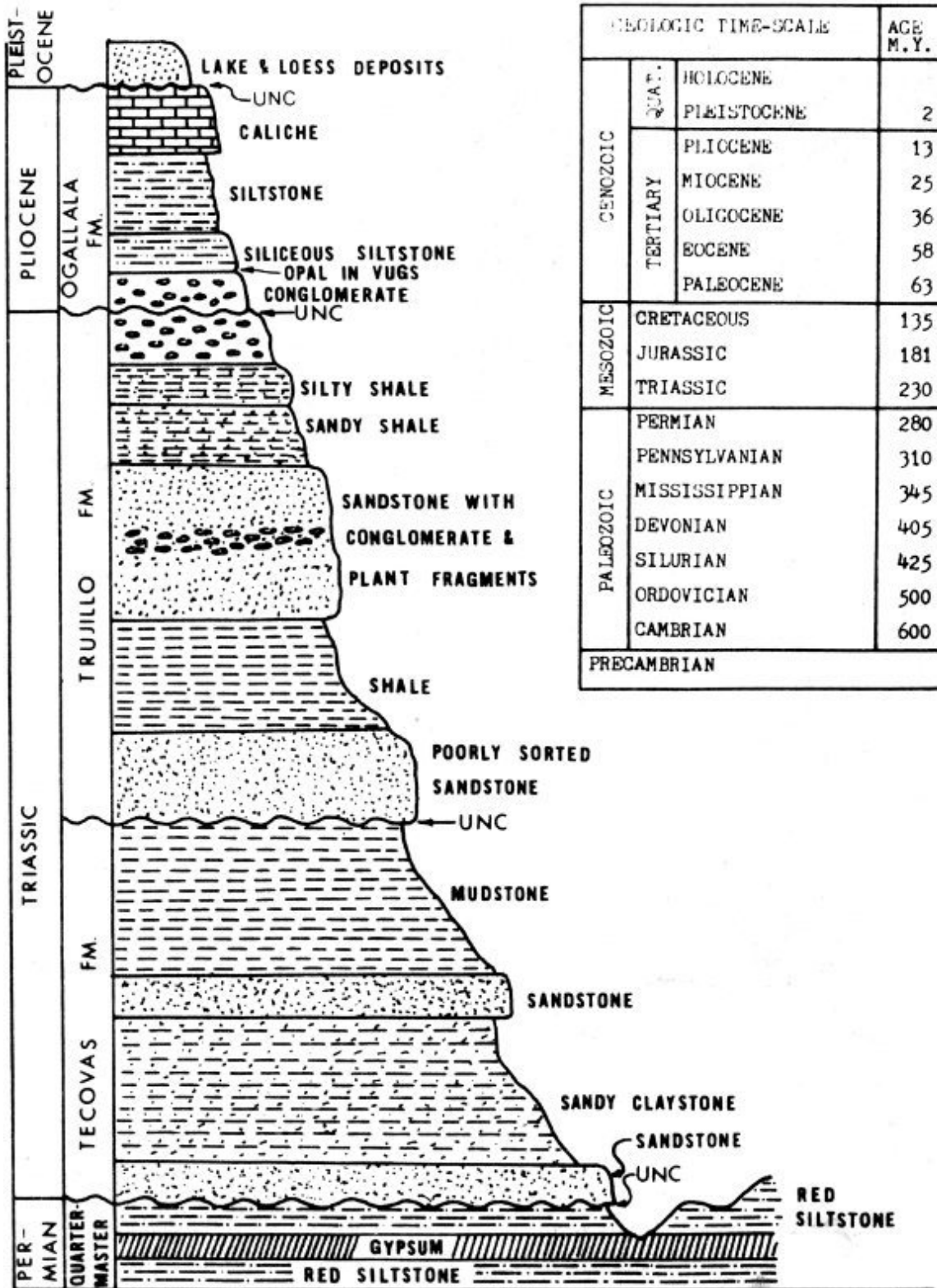


FIGURE 2. STRATIGRAPHIC SECTION AT PALO DURO CANYON WITH GEOLOGIC TIME-SCALE FOR REFERENCE.

GEOLOGIC TIME-SCALE AGE M.Y.

CENOZOIC

QUAT.

HOLOCENE

PLEISTOCENE 2

TERTIARY

PLIOCENE 13

MIOCENE 25

OLIGOCENE 36

EOCENE 58

PALEOCENE 63

MESOZOIC

CRETACEOUS 135

JURASSIC 181

TRIASSIC 230

PALEOZOIC

PERMIAN 280

PENNSYLVANIAN 310

MISSISSIPPIAN 345

DEVONIAN 405

SILURIAN 425

ORDOVICIAN 500

CAMBRIAN 600

PRECAMBRIAN

The lowest and oldest exposed formation in Palo Duro Canyon is the Quartermaster. It was deposited near the edge of a shallow sea that occupied a wide area in Texas some 280 to 230 million years ago ([Figure 3](#)). This was during the Permian Period, the last period of the Paleozoic Era. The sediments in the park area were carried from the east and deposited in a nearshore environment. Sedimentary structures, such as crossbeds and ripplemarks, are present throughout the Quartermaster Formation. Halite casts suggest that there was a high rate of evaporation as sedimentation

occurred. Gypsum (altered anhydrite) is also interpreted to be an evaporite deposit. The gypsum is now seen as horizontal white layers of alabaster and satin-spar varieties within the Quartermaster Formation.

The Quartermaster Formation is mostly siltstone and shale, and is commonly a distinctive red color. This red color is the result of combining oxygen from the air with the iron in the sediments (oxidation) much as a nail rusts after it has been exposed for a long period. The bedded gray zones represent times when there was enough fresh water from the land to offset temporarily the oxidation process. Smaller circular gray areas have organic nuclei that produced local areas of chemically altered iron by a process called reduction.

At the close of the Permian Period and the Paleozoic Era, the Panhandle region was uplifted and a period of widespread erosion followed. Consequently there are no sediments in this area to represent the early or middle portions of the Triassic Period. Breaks, such as this, in the sedimentary record are called UNCONFORMITIES. They may have been caused by a lack of deposition in the area or by an interval during which erosion removed earlier sediments.

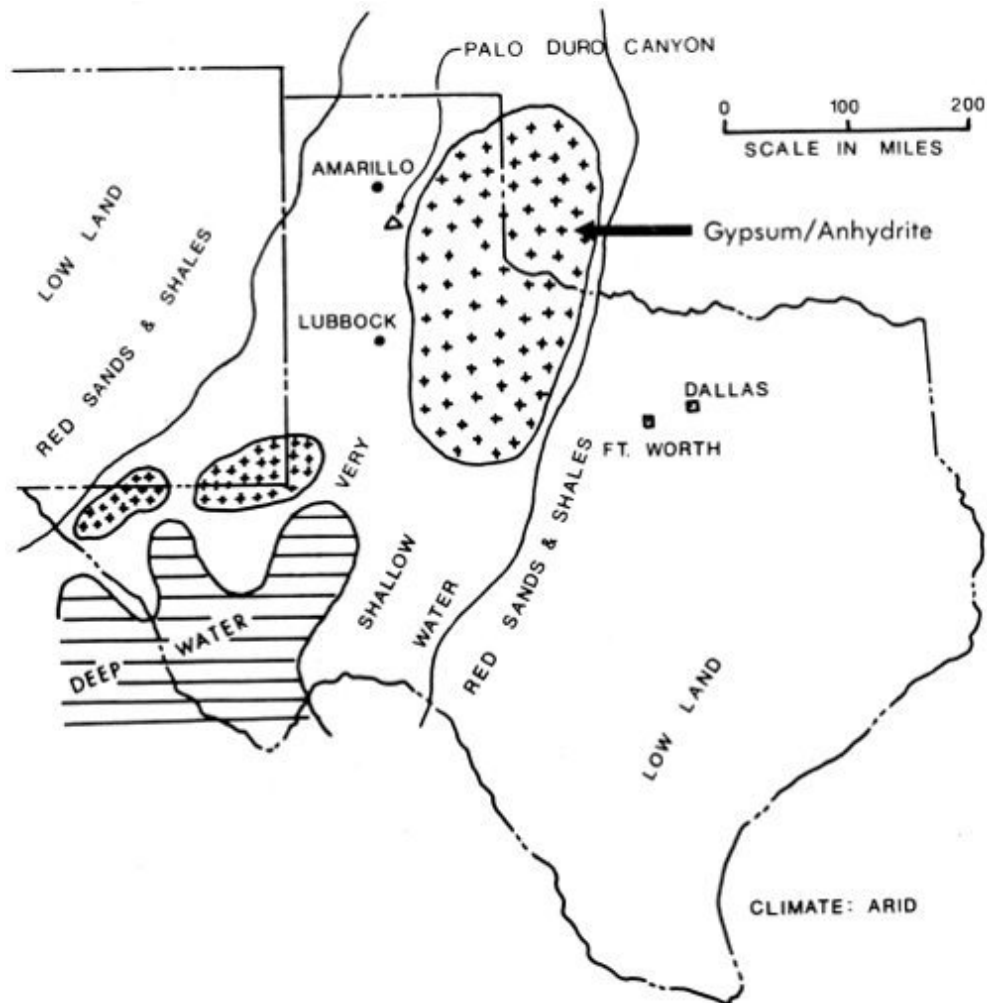


FIGURE 3. PALEOGEOGRAPHIC MAP OF THE PERMIAN PERIOD (240 M.Y.)

The Tecovas Formation was deposited in swamps, lakes, and streams approximately 200 million years ago during the Late Triassic (Figure 4). The Tecovas is mostly purplish lavender, yellow, orange, and buff siltstone and shale. The bright-colored shale of the Tecovas Formation is easily followed for many miles. Amarillo (the Spanish word for yellow) got its name from Amarillo Creek where the yellow bed of the Tecovas Formation crops out far from Palo Duro Canyon. The shale of the Tecovas forms the less steep portions of the canyon walls and often is covered by talus (weathered, broken rock) or vegetation. The uppermost Tecovas is usually mantled with boulders from the overlying sandstone of the Trujillo Formation.

The Tecovas Formation contains numerous concretions or irregularly shaped, weathered rocks. The unusual shape of a concretion is the result of the hardening of the sediments around a nucleus. As the rock weathers, the resistant material surrounding the nucleus remains. Most of the concretions are composed of limonite, hematite, manganite or calcite. Some of the calcite concretions are a variety termed “septarian.” These concretions have calcite ridges in a honeycomb pattern throughout the rock. Some of the concretions are simply nodular or spherical aggregates. Also in the Tecovas, geodes filled or lined internally with calcite crystals are found.

11

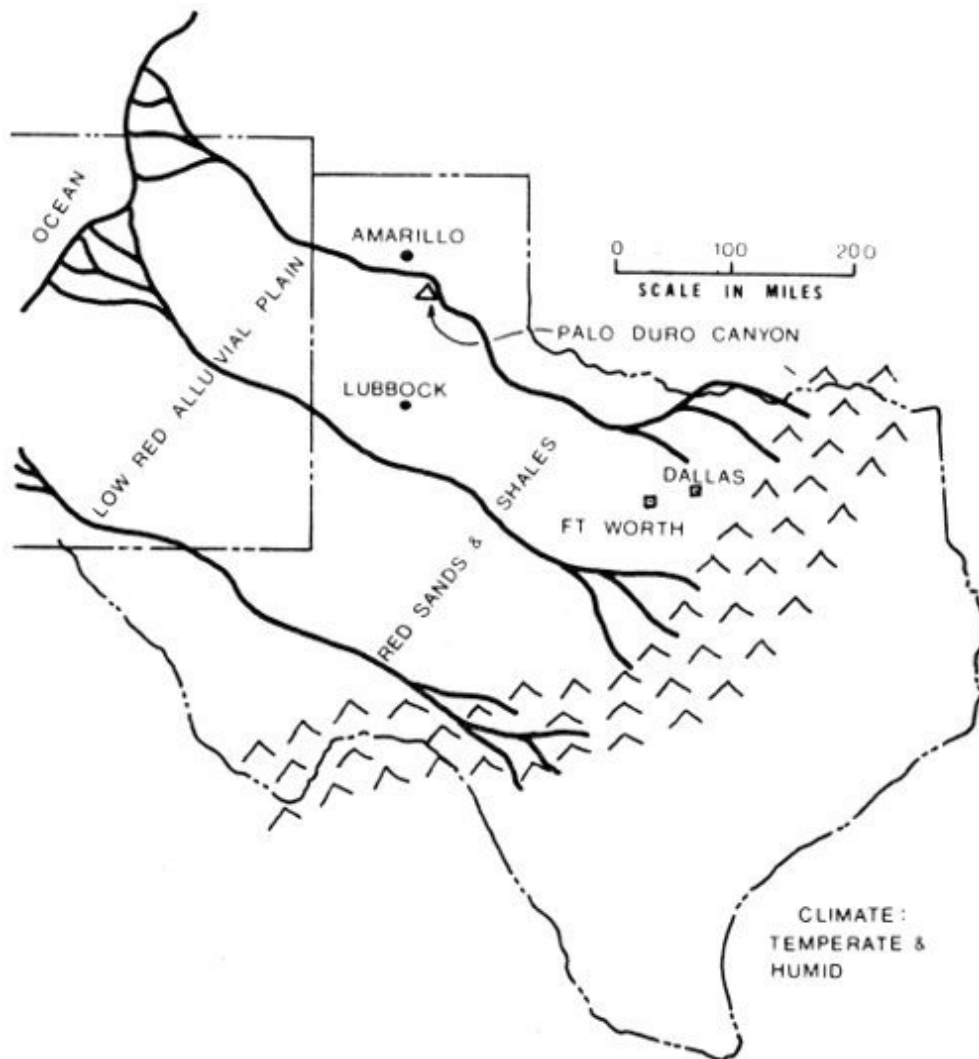


FIGURE 4. PALEOGEOGRAPHIC MAP OF THE TRIASSIC PERIOD (181 M.Y.)

Overlying the Tecovas is the Trujillo Formation. It was deposited by streams that probably originated in an ancient highland southeast of the present Panhandle. These streams were flowing more than 181 million years ago. The sandstone contains some alternating layers of shale and marl-pebble conglomerate. The Trujillo Formation is a resistant formation and forms some of the upper portions of the canyon walls. The steep portions are, in part, the result of a persistent fracture system common in the Trujillo Formation. The sandstone and conglomerate of the Trujillo characteristically exhibit well-developed crossbedding. Their gray color is sometimes obscured by a crust of red mud or iron stain. The uppermost red shale contains mineralized wood. The gray micaceous sandstone has many round sandstone concretions. Some of these are septarian concretions with cracks filled by calcite. Others may contain leaf imprints. 12

There is no evidence that Jurassic sediments were ever deposited in the region. Cretaceous rocks are also missing in this area although water-worn fossil oysters occur in the gravel at the base of the overlying Ogallala. These fossils indicate that marine Cretaceous sediments were deposited nearby and possibly covered the Triassic deposits in the region. The rocks were then eroded away some time between the end of the Cretaceous Period and the beginning of the Pliocene Epoch, a span of about 50 million years.

During the Pliocene Epoch, approximately 2-10 million years ago, the Rocky Mountains were again uplifted. Sediments in streams and floodplains were deposited on the erosional surface of the Trujillo Formation ([Figure 5](#)). These stream-deposited sediments are the Ogallala Formation.

The Ogallala Formation, which forms the upper part of the sequence of rocks exposed in the canyon, is present throughout most of the Panhandle. The formation is important as it is the principal aquifer of the Panhandle and supplies many farms and cities in the region with water. The Ogallala is a siltstone and sandstone that has, in places, been cemented by silica which came from groundwater. The formation contains many pockets of common opal and the basal part is in many places almost a chert. There are also some thin gray shale lenses.

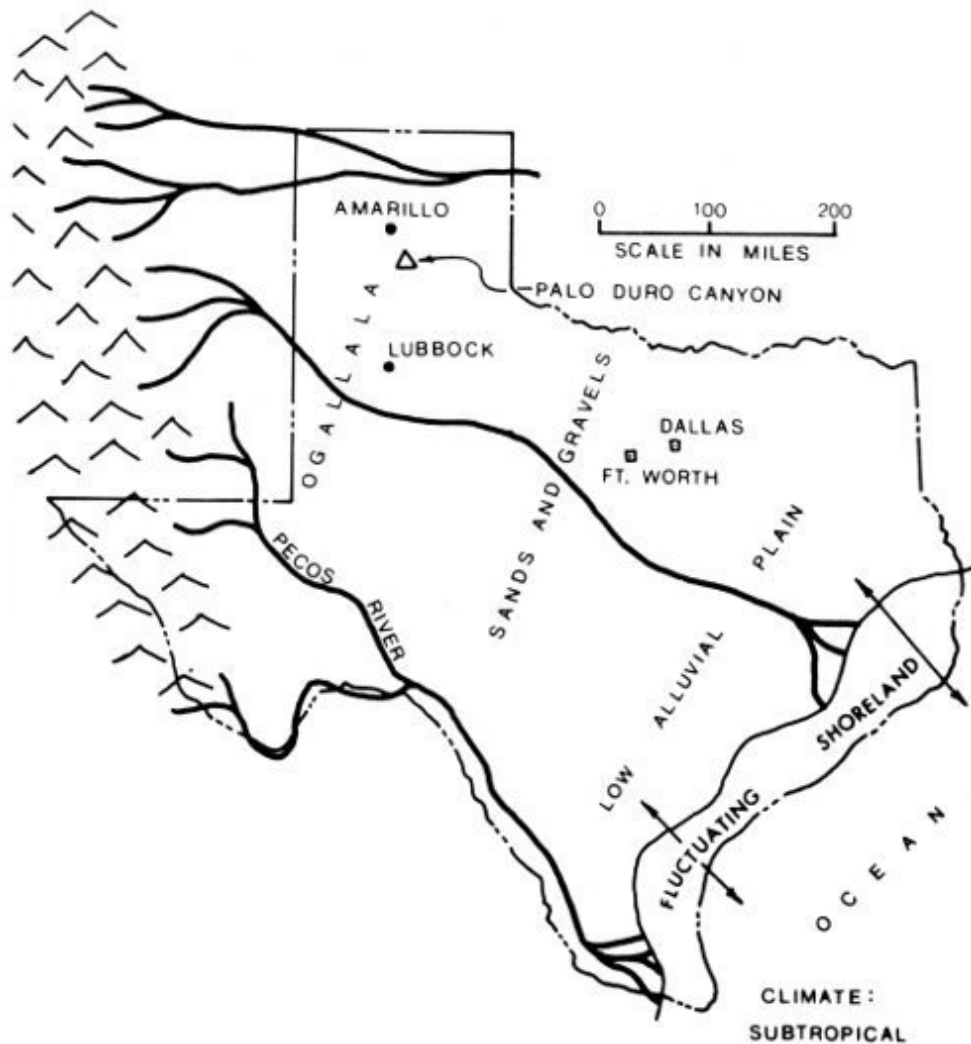


FIGURE 5. PALEOGEOGRAPHIC MAP OF THE PLIOCENE (10 M.Y.)

Scattered over the Ogallala are Late Pliocene and Pleistocene playa lake deposits up to 3 million years old. Some of these are fresh water lake deposits of silt, limestone, and wind-transported sediments or loess. Below these sediments is a layer of caliche which was deposited by evaporation of groundwater rich in calcium carbonate during Late Pliocene and Pleistocene time.

Less than one million years ago, during the Pleistocene Epoch of the Quaternary Period, the Prairie Dog Town Fork of the Red River began eroding headward into the Llano Estacado ([Figure 6](#)). The caprock escarpment is the result of differing resistance to erosion. The faster erosion

of softer layers under the more resistant Ogallala and Trujillo formations forms the steep slopes of the escarpment.

The Prairie Dog Town Fork of the Red River is the small stream flowing in Palo Duro Canyon. Throughout the past million years it has been slowly excavating the canyon. The rate of change has been slow but continuous, carving the steep colorful walls of Palo Duro Canyon, an area of geologic interest and great scenic beauty.

PALEONTOLOGY

During the Permian Period the area that is now Palo Duro Canyon State Park, was a nearly-flat land surface along the edge of a restricted sea. The scarcity of fossils in the Quartermaster Formation indicates that plant and animal life was sparse. The environment was probably unsuited for plant life. It is thought that groundwater near the surface evaporated, leaving large amounts of salt as a residue. Since plants could not grow, animals would not have frequented the area either.

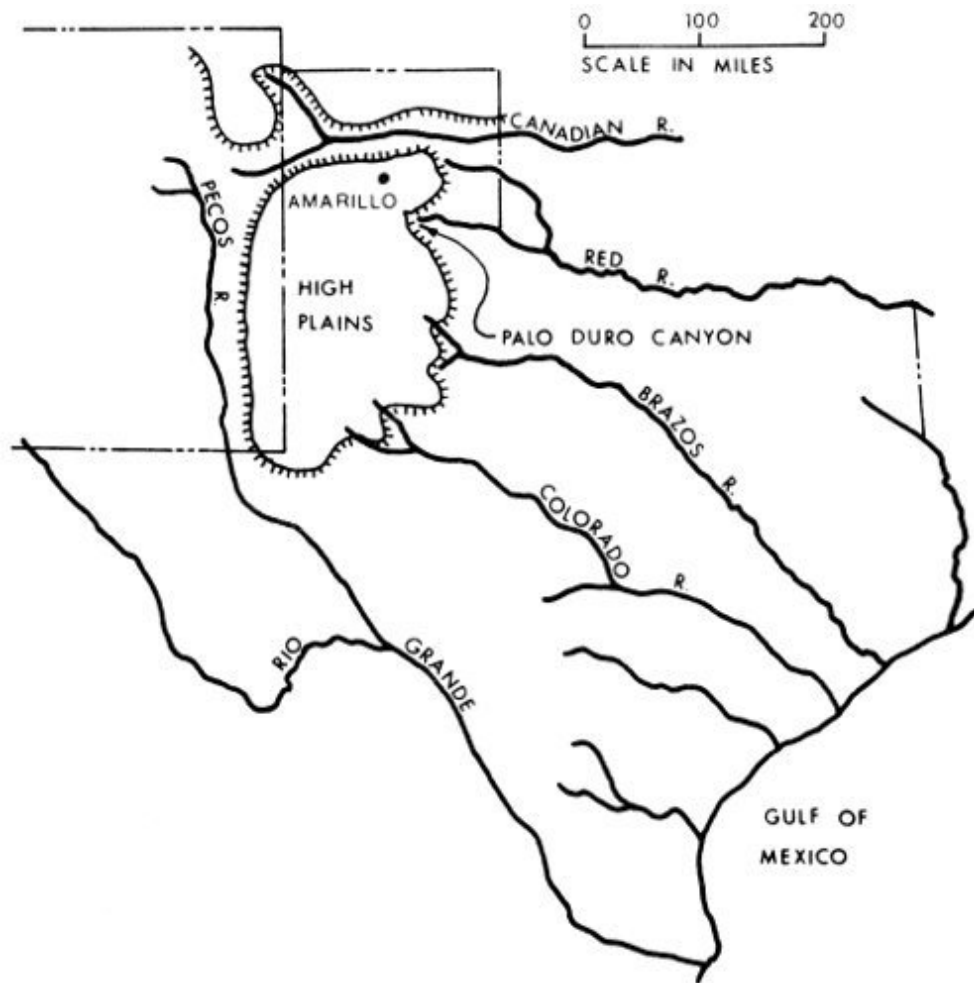


FIGURE 6. HEADWARD EROSION BY THE PECOS, COLORADO, BRAZOS, RED AND CANADIAN RIVERS ISOLATE THE HIGH PLAINS BY THE END OF THE PLEISTOCENE (10,000 YEARS AGO) AND CUT PALO DURO CANYON.

Fossils and rocks of the Tecovas Formation indicate that the sediments were deposited in a swamp and stream environment ([Figure 7](#)). As time went on and the land continued rising, the climate became drier and some of the earlier life forms disappeared.

Remains of *Metoposaurus* ('*Buettneria*'), the last of a long line of giant amphibians, are found here. These animals lived in large ponds. *Metoposaurus* buried themselves in the bottom of a pond and waited for fish to pass. With the aid of a third eye in the middle of its head, the animal could direct its huge mouth to its prey. *Metoposaurus* was so bulky that it is thought that it did not leave the water because its weak legs could not support its weight on land.

Living in shallower areas of the swamp were a group of semiaquatic reptiles known as phytosaurs. Phytosaurs looked very much like giant crocodiles with a nostril on the top of their heads, which permitted them to lie submerged just below the surface of the water. They probably fed on fish and smaller reptiles. Phytosaurs reached a length of 50 feet.

A heavily armored aetosaur, *Desmosuchus* also lived in the Park area during the Triassic. They attained a length of about 10 feet. These reptiles probably were herbivorous (plant eating). A unique feature of *Desmosuchus* was a fringe of backward-pointing horns around their necks. These possibly served to protect them from the carnivorous (meat eating) phytosaurs.

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Figure 7. Life of Late Triassic time, showing restorations of the animals and plants that are now found as fossils in the Chinle beds of New Mexico and Arizona. In the water is the gigantic labyrinthodont amphibian, *Eupelor*, an animal some six feet or more in length. Lying on the bank is the crocodile-like thecodont reptile, *Phytosaurus*, large individuals of which may be twenty or thirty feet long. Behind the phytosaur, in the distance, is the armored thecodont, *Desmosuchus*, ten feet long, and in the foreground is the small, bipedal thecodont, *Hesperosuchus*. In the left background are two individuals of the early saurischian dinosaur, *Coelophysis*, reptiles about ten feet

in length. These animals lived in a tropical environment of moderate topography, crossed by many sluggish rivers and dotted with lakes. Numerous volcanoes rose above the general level of the land. Large, araucarian trees were abundant, stout scouring rushes or horsetails ten or fifteen feet high were everywhere, and the ground was covered with abundant ferns.

Also found in the Tecovas and Trujillo formations are fossil lung-fish teeth. Lung-fish are a type of fish that can breathe air, enabling them to move from pond to pond. Footprints of a chicken-sized dinosaur have also been found. The Middle Triassic flora was dominated by giant palm-like trees. Also found are remains of a few large ferns and horsetails. As the climate became drier and the swamps began to disappear, coniferous (evergreen) trees such as *Araucarioxylon* became plentiful. These trees can be found in the canyon today as petrified wood. 18

The Ogallala in the park contains very few fossils. A giant tortoise was found near the bend where the road begins to descend into the canyon. Fossil seeds may be seen in the exposure of the Ogallala near the Coronado Lodge.

More extensive Late Pliocene fossil beds are exposed south of the park in Cita Canyon. These beds are younger than the Ogallala and are stream and basin deposits. The fauna and flora found here suggest a broad, flat, grassy plain much like the present landscape ([Figure 8](#)). Remains of mastodons, large, elephant-like animals with long upper tusks that were used to dig up roots, are found here. Saber-tooth cats, also present, preyed upon the mastodons. The remains of these, as well as bones of camels, pony-sized horses, and sloths 10 feet high have been found in the vicinity of the canyon. Some of these animals are thought to have lived in the Panhandle a mere 10,000 years ago.

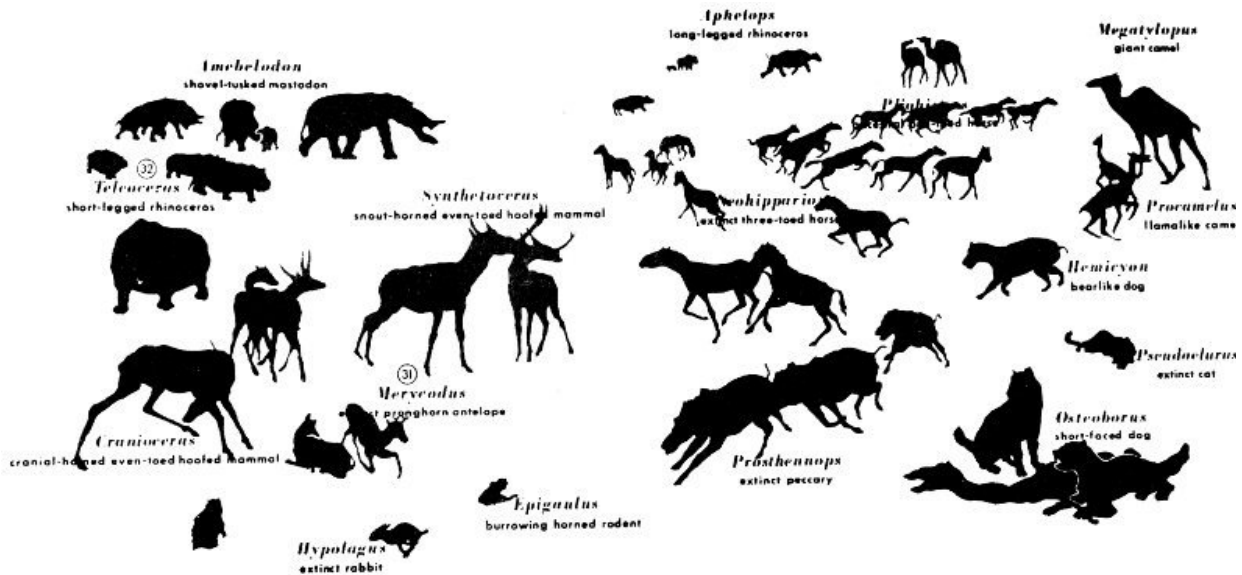


Figure 8.

Amebelodon: shovel-tusked mastodon

Teleoceras: short-legged rhinoceros

Synthetoceras: snout-horned even-toed hoofed mammal

Cranioceras: cranial-horned even-toed hoofed mammal

Merycodus: extinct pronghorn antelope

Hypolagus: extinct rabbit

Epigaulus: burrowing horned rodent

Aphelops: long-legged rhinoceros

Prosthennops: extinct peccary

Osteoborus: short-faced dog

Pseudaelurus: extinct cat

Hemicyon: bearlike dog

Procamelus: llamalike camel

Megatylopus: giant camel

Pliohippus: ancestral one-toed horse

Neohipparion: extinct three-toed horse



FIGURE 22.41 Pliocene Mammals. Early Pliocene life of the southern High Plains. (Mural by J. H. Matternes, courtesy U.S. National Museum.)

Due to limited outcrops in the Canyon proper, Pleistocene fossils are very rare. An excellent collection of fossils from Palo Duro Canyon and the Panhandle area is on display at the Panhandle-Plains Historical Museum. 21



Spectacular Palo Duro, “grand canyon” of the Panhandle Plains, is an exciting experience. One of the nation’s most magnificent scenic attractions, it provides delightful drives and opportunities for hiking, horseback riding and camping in season. Visit Palo Duro and enjoy it.

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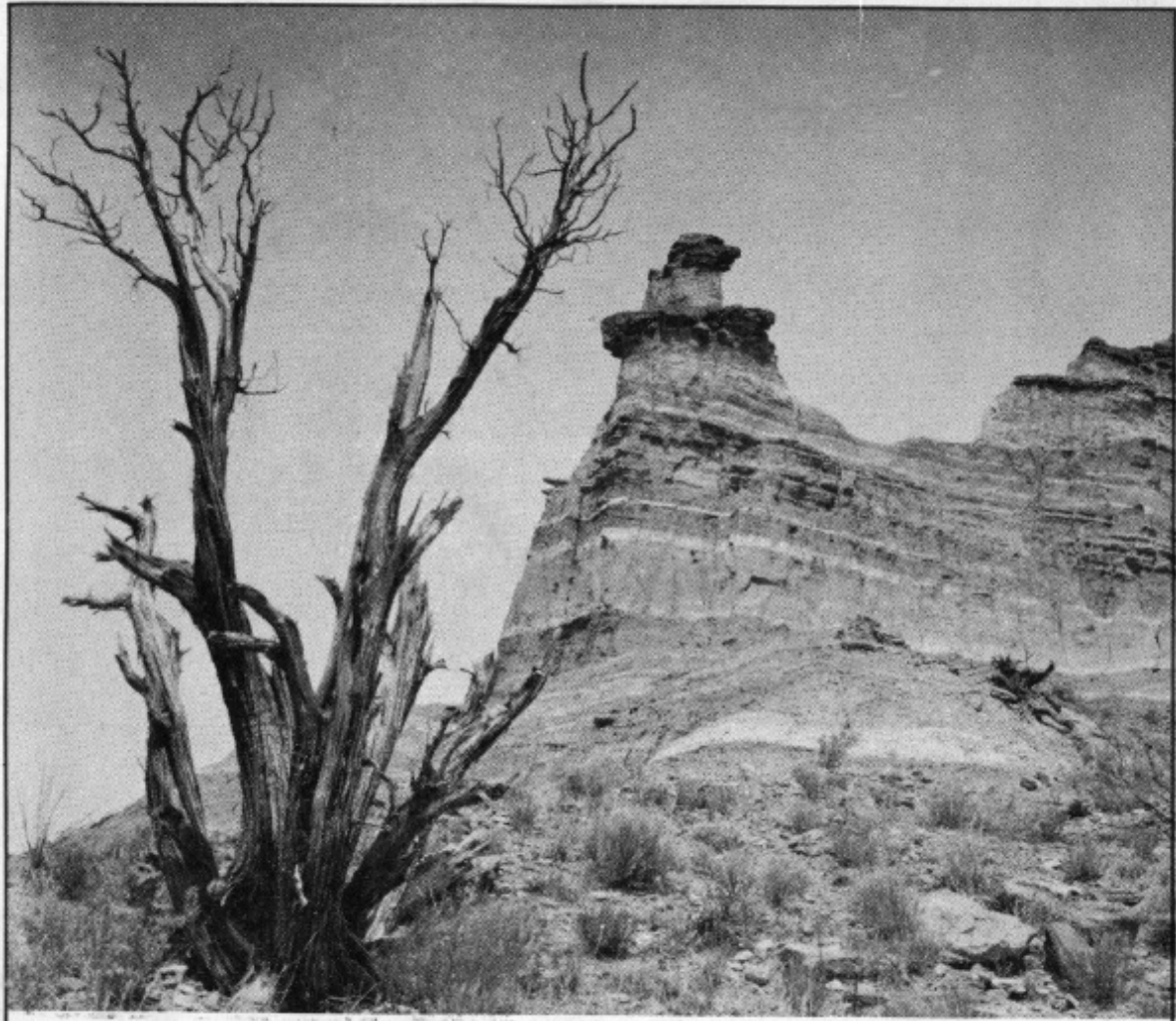
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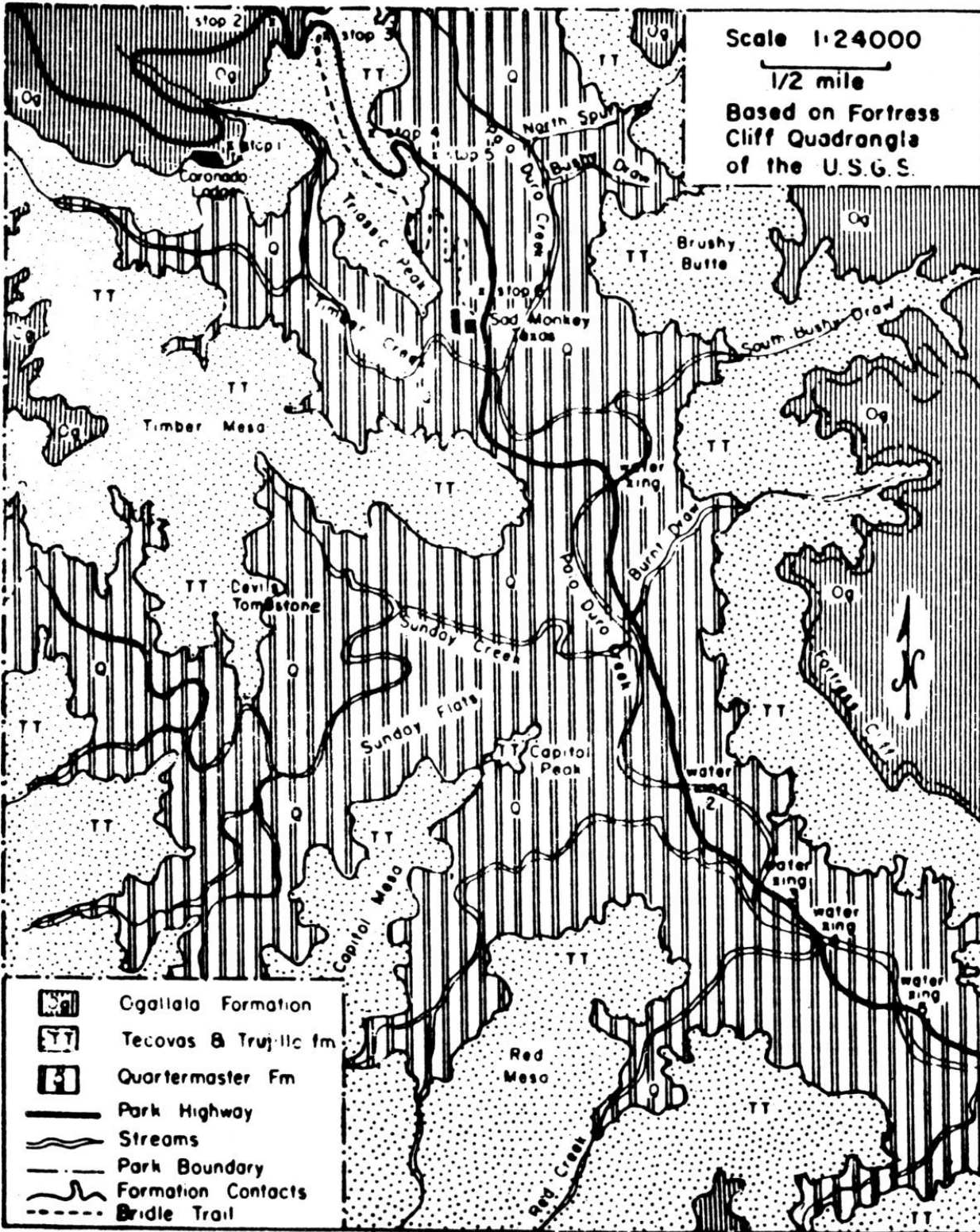
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